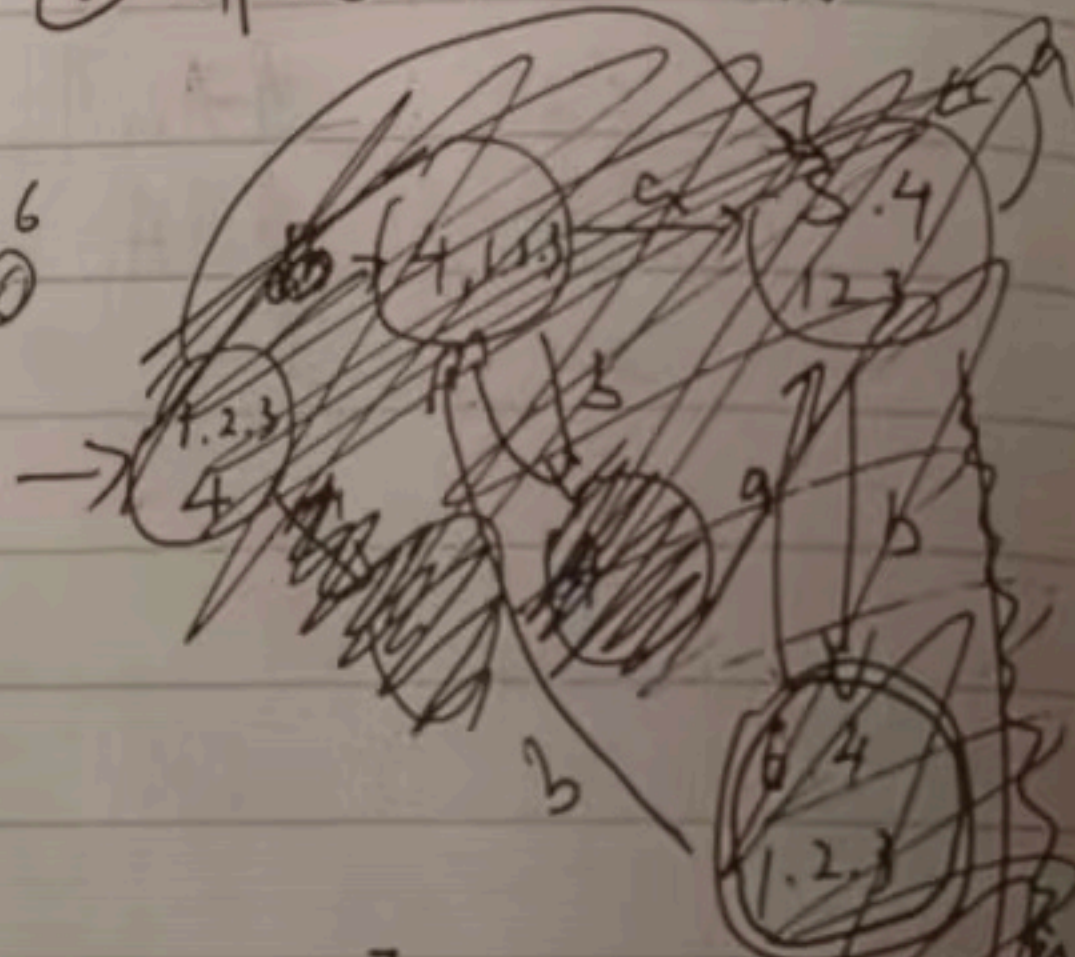
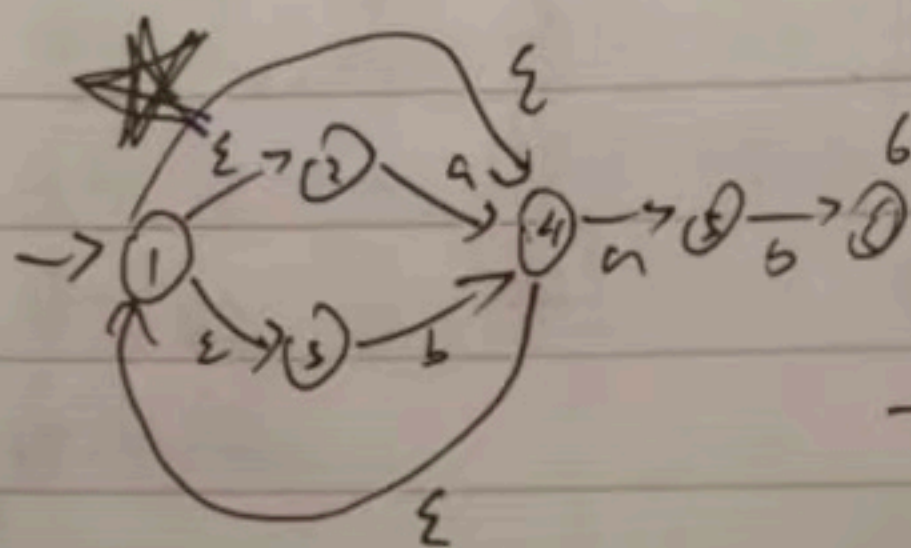


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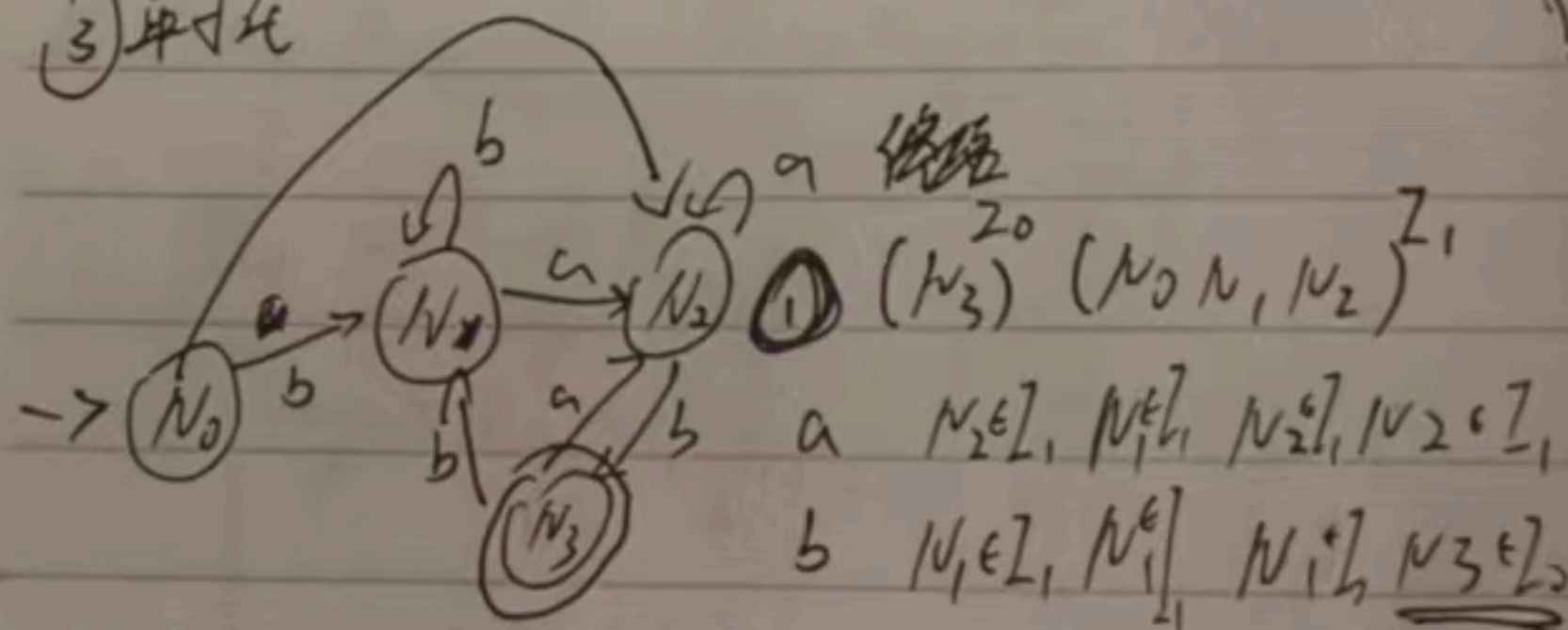
=, 1.A 2.D 3.C 4. ~~D~~ 5.B 6.B 7. ~~A~~ B
 8. ~~A~~ B 9.D 10.C

二, Ca^{2+} 的抵抗 $17A$.

② 消去 $\sim$ $\neg A \Rightarrow \neg A$



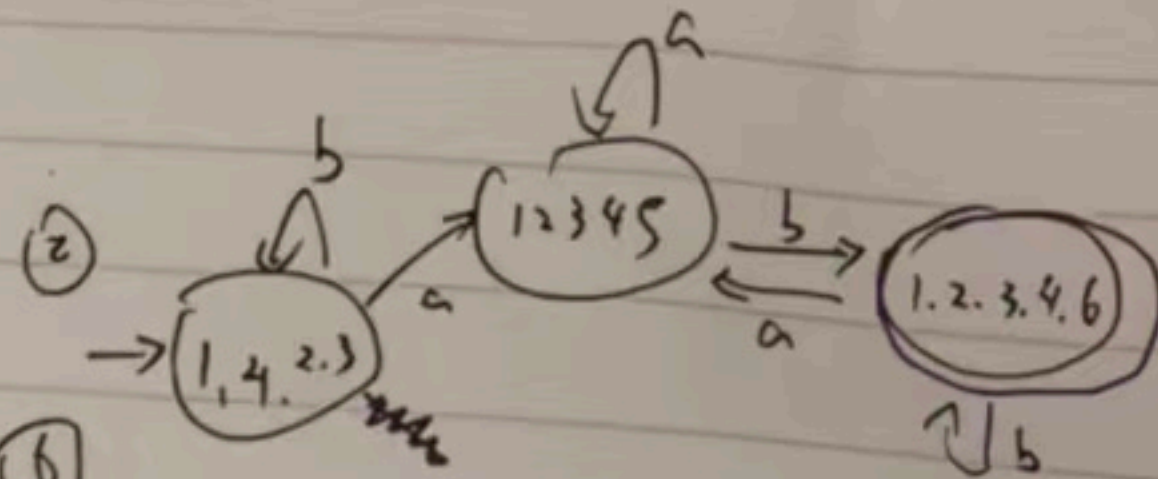
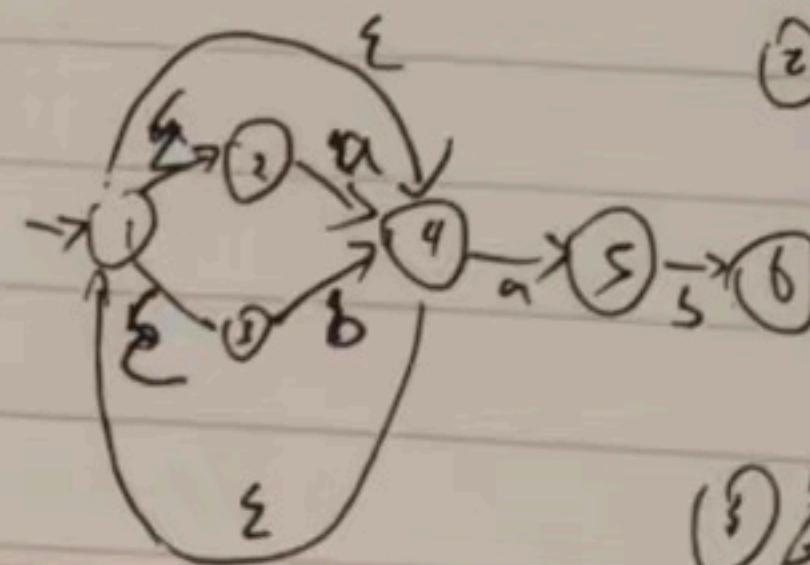
③ 单位



② 见 I. 和 I' 的地质剖面!

	$(N_3) Z_3$	$(N_3 N_1) Z_1$	$(N_2) Z_2$
a	$N_2 \in Z_1$	Z_1	Z_3
b	Z_1	Z_1	Z_3

④



③ 子午月

17] : $G: S \rightarrow a/n/ (7)$
 $7 \rightarrow 7, s/s.$

(1) 改 G' :

$$\begin{cases} S \rightarrow a | \Lambda | (T) \\ T \rightarrow ST' \\ T' \rightarrow , ST' | \epsilon \end{cases}$$

② 是. ① 无互递归 ② $72RS7$ 不交 ③ $72RS7$ $72RS7$ 不交

(3) 预测分析: $a \wedge () , \#$

$S \rightarrow a \mid S \rightarrow \Lambda \mid S \rightarrow (S)$ $T \rightarrow ST' \mid T \rightarrow ST' \mid T \rightarrow ST'$ $T' \rightarrow \epsilon \mid T' \rightarrow , ST'$	$S \rightarrow a \mid S \rightarrow \Lambda \mid S \rightarrow (S)$ $T \rightarrow ST' \mid T \rightarrow ST' \mid T \rightarrow ST'$ $T' \rightarrow \epsilon \mid T' \rightarrow , ST'$
---	---

$A \rightarrow a$ 填入 Follow
 $A \rightarrow \epsilon$ 填入 Follow

	S	T	T'
Follow	$\{a, \Lambda, (,)\}$	$\{a, \Lambda, (,)\}$	$\{, , \epsilon\}$
Follow	$\{\#, ,\}$	$\{>\}$	$\{>\}$

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100. j relop a. b. 102.

101 j. - , - , ~~102~~

102 jnz A , - 106

714 ~~102~~
dals

103 j. - , - 104

104 jnz B , - 106

105 j. - , - 109.

106 + ~~107~~ , T₁ , y. Z.

~~107 = , T₁ , - , X 108. j. - , - 112~~

107 = , T₁ , - , X 108. j. - , - 112

109. - , T₂ , y. Z

110 = , T₂ , - , X.

111 j. - - , 102

112 112

112:

五. (同21-1) 是奇数, 完全相同)

六. 给出中间代码.

(1) $G: \begin{cases} T = A - B \\ U = A - C \\ V = T + U \\ W = V + U \\ Y = T - W \end{cases}$ 可用 R_0, R_1 寄存器 W, Y

七. 反向为块处理 (x, y) 表示 (待用, 方块)

(5) $Y = T - W$ (1, y) (1, 1) (1, 1)

(4) $W = V + U$ (5, y) (1, 1) (1, 1)

(3) $V = T + U$ (4, y) (5, y) (4, y)

(2) $U = A - C$ (3, y) (1, 1) (1, 1)

(1) $T = A - B$ (5, y) (2, y) (1, 1)

过程: $A(1, 1) \Rightarrow (2, y) \Rightarrow (1, y)$

$B(1, 1) \Rightarrow (1, y)$

$C(1, 1) \Rightarrow (2, y)$

$T(1, 1) \Rightarrow (5, y) \Rightarrow (3, y) \Rightarrow (1, 1)$

$U(1, 1) \Rightarrow (4, y) \Rightarrow (3, y) \Rightarrow (1, 1)$

$V(1, 1) \Rightarrow (4, y) \Rightarrow (1, 1)$

$W(1, y) \Rightarrow (5, y) \Rightarrow (1, 1)$

$Y(1, y) \Rightarrow (1, 1)$

先与到图式
再更改状态

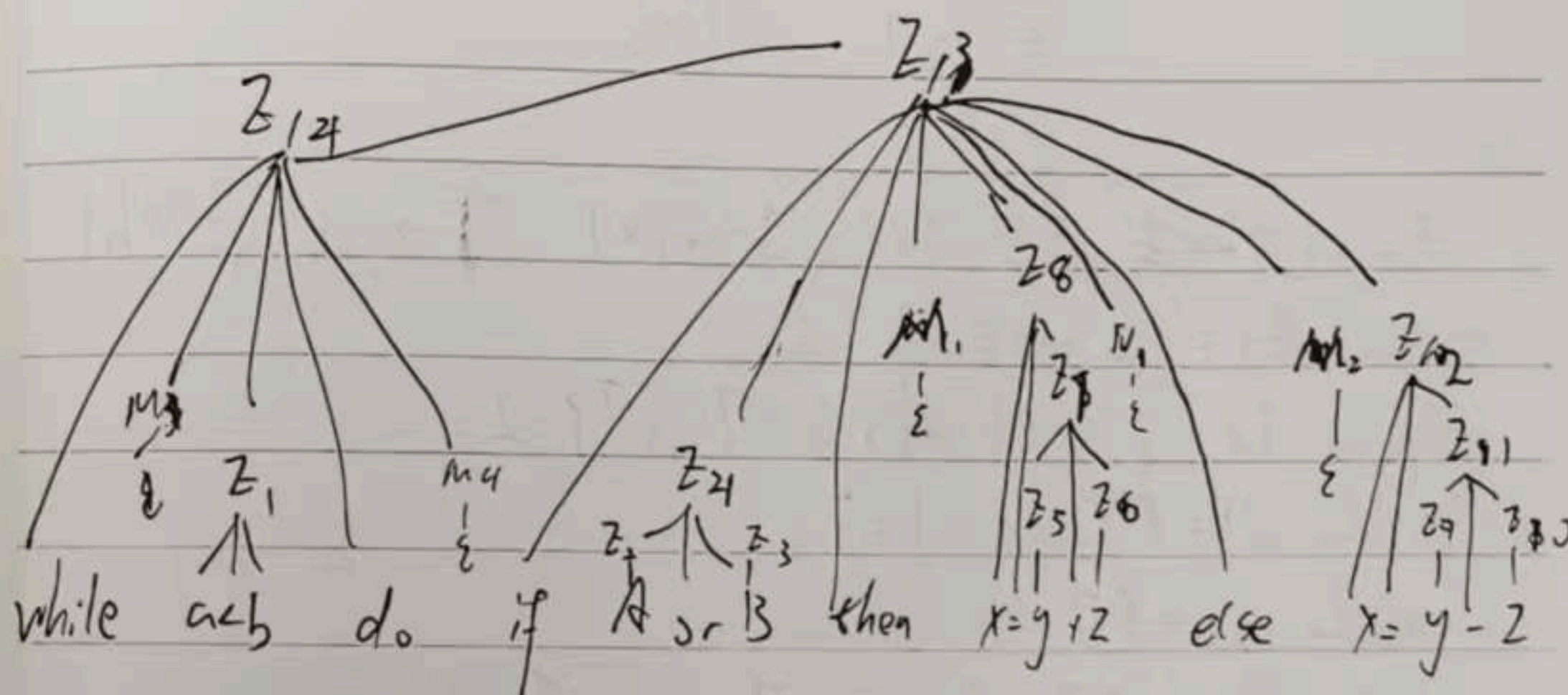
11月20日

(2) 生成中间代码, 使用两个 Register

Op	Op	Op	Rval	Aval
$T = A - B$	LD R_0, A		$\{R_0\} = A$	$\{T\} = R_0$
	SUB R_0, B			
$U = A - C$	LD R_1, A		$\{R_1\} = U[R_0] = T$	$\{U\} = R_1$
	SUB R_1, C			
$V = T + U$	ST R_0, T		$\{R_0\} = V, \{R_1\} = U$	$\{V\} = R_0$
	ADD R_0, U			$\{T\} = R_1$
$W = V + U$	ADD R_0, U		$\{R_0\} = W$	
$Y = T - W$	LD R_1, T		$\{R_1\} = Y$	
	SUB R_1, R_0		$\{R_0\} = W$	
	ST R_1, Y			
	ST R_0, W			

七. 控制流图的中代码翻译; 这里用 one-pass 的交点

流程图:



100/ j. rep. op. a. b. 0
101/ j. - 0
102/ j. 2. A. - 0
103/ j. - 0

★ 这里并语中
Z-7id 不需要emit

八. $G: \begin{cases} Z \rightarrow T^+T \mid T^*T \\ T \rightarrow a \mid b \end{cases}$

(1) 闭包法: $\begin{cases} Z' \rightarrow Z & ① \\ Z \rightarrow T^+T \mid T^*T & ② \text{ ③} \\ T \rightarrow a \mid b & ④ \text{ ⑤} \end{cases}$

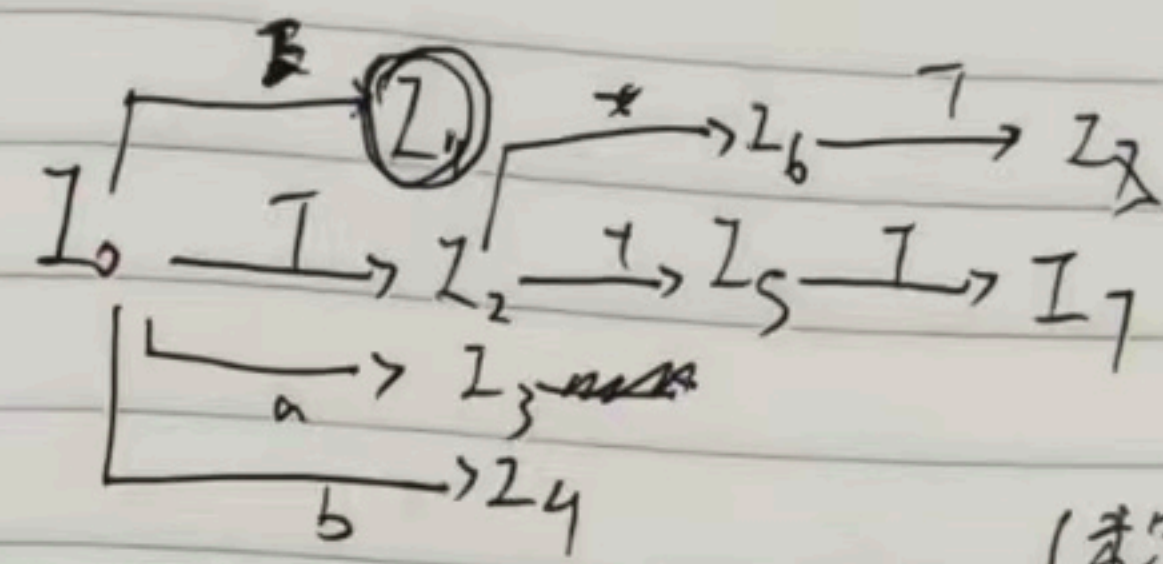
LR(0) 项目:

$Z' \rightarrow \cdot Z$	$Z \rightarrow \cdot T^+T$	$Z \rightarrow \cdot T^*T$	$T \rightarrow \cdot a$	$T \rightarrow \cdot b$
$Z' \rightarrow Z \cdot$	$Z \rightarrow T \cdot T^+T$	$Z \rightarrow T \cdot T^*T$	$T \rightarrow a \cdot$	$T \rightarrow b \cdot$
$Z \rightarrow \cdot T^+T$	$Z \rightarrow T \cdot T^+T$	$Z \rightarrow \cdot T^*T$		
$Z \rightarrow T \cdot T^+T$	$Z \rightarrow T T \cdot T^+T$	$Z \rightarrow \cdot T^*T$		
$Z \rightarrow T T \cdot T^+T$	$Z \rightarrow T T T \cdot T^+T$	$Z \rightarrow \cdot T^*T$		
$Z \rightarrow T T T \cdot T^+T$	$Z \rightarrow T T T T \cdot T^+T$	$Z \rightarrow \cdot T^*T$		

(2)

$Z_0 = \{Z' \rightarrow \cdot Z, Z \rightarrow \cdot T^+T, Z \rightarrow \cdot T^*T, T \rightarrow \cdot a, T \rightarrow \cdot b\}$
 $Go(Z_0, Z) = \{Z' \rightarrow Z \cdot\} = Z_1$
 $Go(Z_0, T) = \{Z \rightarrow T \cdot T^+T, Z \rightarrow T \cdot T^*T\} = Z_2$
 $Go(Z_0, a) = \{T \rightarrow a \cdot\} = Z_3$
 $Go(Z_0, b) = \{T \rightarrow b \cdot\} = Z_4$
 $Go(Z_2, T) = \{Z \rightarrow T T \cdot T^+T, Z \rightarrow T T \cdot T^*T\} = Z_5$
 $Go(Z_2, *) = \{Z \rightarrow T T^* \cdot T, T \rightarrow \cdot a, T \rightarrow \cdot b\} = Z_6$
 $Go(Z_5, T) = \{Z \rightarrow T T T \cdot T^+T\} = Z_7, Go(Z_5, a) = Z_3, Go(Z_5, b) = Z_4$
 $Go(Z_6, T) = \{Z \rightarrow T T^* T \cdot\} = Z_8, Go(Z_6, a) = Z_3, Go(Z_6, b) = Z_4$

DFA:



没有对 $Z' \rightarrow Z \cdot$ 的闭包!

(未写出的闭包项未入图)

(3) LR(0) 分析表:

ACTION.						GOTO		
	+	x	a	b	#	Z'	Z	T
Z ₀			S ₃	S ₄			Z ₁	Z ₂
Z ₁					acc			
Z ₂	S ₅	S ₆						
Z ₃	r ₄	r ₄	r ₄	r ₄	r ₄			
Z ₄	r ₅	r ₅	r ₅	r ₅	r ₅			
Z ₅			S ₃	S ₄				Z ₇
Z ₆			S ₃	S ₄				Z ₈
Z ₇	r ₂	r ₂	r ₂	r ₂	r ₂			
Z ₈	r ₃	r ₃	r ₃	r ₃	r ₃			

(4) 无冲突!